LESSON 12 · CHEAT SHEET

The Pay Gap Report

THE SIX FIGURES (UK DEFINITION)

Hourly rate	`annual_salary / 52 / weekly_hours`
Any gap	`(men - women) / men * 100`
Mean / median hourly gap	on `hourly_rate`, everyone
Mean / median bonus gap	on `bonus`, only `bonus > 0`
% receiving a bonus	with bonus / **all** of that gender
Quartiles	`qcut` hourly into 4, % women per band

PROJECT SHAPE

```
my_work/paygap.py      # functions only, no print
my_work/check1.py      # import paygap; print(...)
my_work/report.py      # runs it all, writes files
python my_work/check1.py   # from the lesson folder
```

`import paygap` works because the script and the module sit in the same folder.

A FUNCTION WITH A DOCSTRING

```
def gap_percent(men, women):
    """UK definition, 1 decimal, plain float."""
    return float(round((men - women) / men * 100, 1))
```

PICKING GROUPS

```
men = df.loc[df["gender"] == "M", "hourly_rate"]
women = df.loc[df["gender"] == "F", "hourly_rate"]
paid = df[df["bonus"] > 0]
men.mean()   men.median()   len(men)
(df["gender"] == "F").sum()        # count women
```

QUARTILES

```
labels = ["Lower", "Lower middle",
          "Upper middle", "Upper"]
bands = pd.qcut(df["hourly_rate"], 4, labels=labels)
t = pd.crosstab(bands, df["gender"],
                normalize="index") * 100
t = t.round(1).rename(columns={"F": "pct_women",
                               "M": "pct_men"})
t.index.name = "quartile"
```

PER GROUP, BY HAND

```
rows = []
for dept, g in df.groupby("department"):
    rows.append({"department": dept,
                 "employees": len(g)})
table = (pd.DataFrame(rows)
           .set_index("department")
           .sort_values("employees", ascending=False))
```

DICTS IN, DICTS OUT

```
result = {"mean_gap": 19.7, "median_gap": 18.7}
result["mean_gap"]                  19.7
for key, value in result.items():
    print(f"{key}: {value}")
```

OUTPUT (LESSON 11, CONDENSED)

```
plt.bar(x, women, label="Women")
plt.bar(x, men, bottom=women, label="Men") # stack
plt.savefig(OUT / "quartiles.png", dpi=150)

with pd.ExcelWriter(path) as writer:
    summary.to_excel(writer, sheet_name="Summary",
                     index=False)
navy = PatternFill("solid", fgColor="1F3A5F")
for ws in wb.worksheets:
    for cell in ws[1]:
        cell.font = Font(bold=True, color="FFFFFF")
        cell.fill = navy
    ws.freeze_panes = "A2"
```

FORMATTING THE REPORT

```
f"{value:>5}%"        # right-align in 5
"=" * 48              # a line
round(100 - median_gap)   # pence per £1
```

GOTCHAS

- Divide by **men's** pay, always.
- Bonus gap: only people with a bonus. % receiving: everyone.
- Hourly, never annual.
- `round()` on a pandas number gives `np.float64`; wrap in `float()`.
- Changed `paygap.py`? Re-run the check script; nothing reloads by itself.